

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    accuracy_score,
    confusion_matrix,
    classification_report
)

# =====
# Load Dataset
# =====

df = pd.read_csv("heart.csv")

print("First 5 Rows:")
print(df.head())

print("\nShape of Dataset:")
print(df.shape)

print("\nMissing Values:")
print(df.isnull().sum())

# =====
# Features and Target
# =====

X = df.drop("target", axis=1)
y = df["target"]

# =====
# Train-Test Split
# =====

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)

# =====
# Decision Tree Classifier
# =====

dt = DecisionTreeClassifier(
    max_depth=4,
    random_state=42
)

dt.fit(X_train, y_train)

dt_pred = dt.predict(X_test)

dt_acc = accuracy_score(y_test, dt_pred)

print("\n=====")
print("Decision Tree Results")
print("=====")

print("Accuracy:")
print(dt_acc)

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, dt_pred))

```

```

print("\nClassification Report:")
print(classification_report(y_test, dt_pred))

# =====
# Decision Tree Visualization
# =====

plt.figure(figsize=(20, 10))

plot_tree(
    dt,
    feature_names=X.columns,
    class_names=["No Disease", "Disease"],
    filled=True
)

plt.title("Decision Tree Visualization")
plt.show()

# =====
# Random Forest Classifier
# =====

rf = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

rf.fit(X_train, y_train)

rf_pred = rf.predict(X_test)

rf_acc = accuracy_score(y_test, rf_pred)

print("\n=====")
print("Random Forest Results")
print("=====")

print("Accuracy:")
print(rf_acc)

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, rf_pred))

print("\nClassification Report:")
print(classification_report(y_test, rf_pred))

# =====
# Accuracy Comparison
# =====

plt.figure(figsize=(6, 4))

models = ["Decision Tree", "Random Forest"]
scores = [dt_acc, rf_acc]

plt.bar(models, scores)

plt.ylabel("Accuracy")
plt.xlabel("Models")
plt.title("Decision Tree vs Random Forest Accuracy")

plt.show()

# =====
# Feature Importance
# =====

importance = pd.DataFrame({
    "Feature": X.columns,
    "Importance": rf.feature_importances_
})

```

```
importance = importance.sort_values(
    by="Importance",
    ascending=False
)

print("\nTop 10 Important Features:")
print(importance.head(10))

plt.figure(figsize=(10, 6))

sns.barplot(
    x="Importance",
    y="Feature",
    data=importance.head(10)
)

plt.title("Top 10 Feature Importances")
plt.show()

# =====
# Cross Validation
# =====

cv_scores = cross_val_score(
    rf,
    X,
    y,
    cv=5
)

print("\nCross Validation Scores:")
print(cv_scores)

print("\nAverage Cross Validation Accuracy:")
print(cv_scores.mean())

# =====
# Final Message
# =====

print("\nTask 5 Completed Successfully")
```



First 5 Rows:

```

age  sex  cp  trestbps  chol  fbs  restecg  thalach  exang  oldpeak  slope  \
0    52   1   0      125   212    0         1     168    0       1.0     2
1    53   1   0      140   203    1         0     155    1       3.1     0
2    70   1   0      145   174    0         1     125    1       2.6     0
3    61   1   0      148   203    0         1     161    0       0.0     2
4    62   0   0      138   294    1         1     106    0       1.9     1

```

```

ca  thal  target
0   2     3      0
1   0     3      0
2   0     3      0
3   1     3      0
4   3     2      0

```

Shape of Dataset:  
(1025, 14)

Missing Values:

```

age      0
sex      0
cp       0
trestbps 0
chol     0
fbs      0
restecg  0
thalach  0
exang    0
oldpeak  0
slope    0
ca       0
thal     0
target   0
dtype: int64

```

=====  
Decision Tree Results  
=====

Accuracy:  
0.8

Confusion Matrix:

```
[[71 31]
 [10 93]]
```

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.88      | 0.70   | 0.78     | 102     |
| 1            | 0.75      | 0.90   | 0.82     | 103     |
| accuracy     |           |        | 0.80     | 205     |
| macro avg    | 0.81      | 0.80   | 0.80     | 205     |
| weighted avg | 0.81      | 0.80   | 0.80     | 205     |

Decision Tree Visualization

